a Data curation and cohort construction (ChEMBL37)

ChEMBL37 SQLite archive

1,563,556

Human single-protein activity candidates

Curation & filtering

- ✓ Exact IC_{50} records
- ✓ Traceable provenance
- ✓ Valid structures / units
- ✓ Human single-protein targets

690,683 exact IC_{50} records

Pair construction

Same-target assay matching

153,468 molecule–pair rows

83,381 assay-pair IDs

Filtered repeats

79,810 molecule–pair rows

12,392 supported assay pairs

Evaluation splits

Five scaffold splits

Scaffolds held out

b Modeling framework and tasks

Molecular structure + source potency

Morgan bits
Assay-A pIC_{50}

Measured references

Actual ChEMBL examples

SAGP

Potency-axis mean

Source-potency axis

Chemical residual

Molecular Tanimoto kernel

GP mean + covariance

Shared posterior (schematic)

Reference summaries

Neighborhood similarity
Reference dispersion
Posterior uncertainty

Continuous losses → GP

Detached decision adapters

stop ∇

SAR risk

Task 1

Sensitivity

Magnitude + risk

Risk / magnitude losses → adapters

Task 1

SAR transfer

Task 2

Molecular sensitivity

Training targets

Observed B → labels only

Task 1: analogue transfer

B-gap = $pIC_{50,B(i)} - pIC_{50,B(j)}$

Robust ordering / loss of ordering

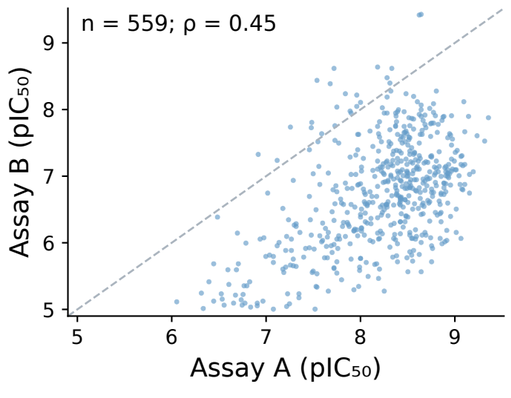
Task 2: molecular sensitivity

r = |(B − A) − training-pair median|

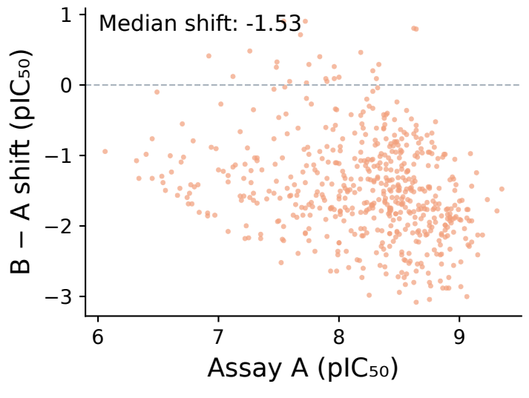
Magnitude + risk at r ≥ 0.5

c Representative data patterns

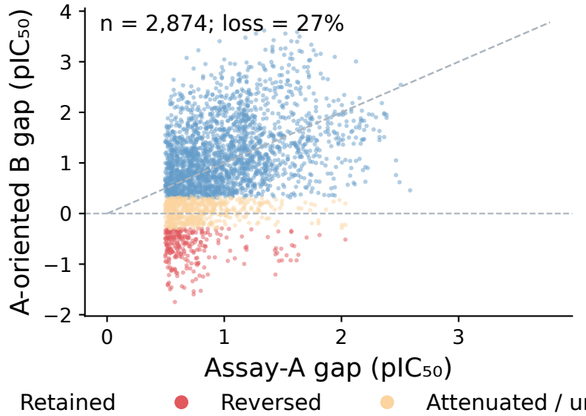
i Assay potency correlation



ii Signed shift (B − A)



iii Analogue agreement



Retained ● Reversed ● Attenuated / uncertain

iv Within-pair support

